

CyberShield Ledger: Academic Notes & Research Summary

A Decentralized, AI-Assisted Cybercrime Intake, Evidence Vault & Triage Platform

2. Related Works & Literature Context

Existing studies and platforms in cybercrime management fall into three main paradigms: Centralized Government Portals, Pure On-Chain Blockchain Loggers, and Standalone NLP Classifiers. CyberShield Ledger unifies their strengths while overcoming critical privacy and scalability flaws.

Approach / Paradigm	Key Characteristics	Critical Limitations	CyberShield Advantage
Centralized Portals (e.g. NCRP 1930, US IC3)	Central database storing citizen reports.	Manual triage delays, no real-time threat graph, vulnerable to insider tamper.	Automated AI Triage & Encrypted Vault (AES-256-GCM + Threat Graph).
Pure On-Chain Loggers (Hyperledger / Ethereum)	Stores complaint narratives & PII directly on smart contracts.	Severe GDPR / Privacy violations, high gas costs, public exposure of PII.	Off-Chain Privacy Hybrid: Only SHA-256 bundle hashes on-chain.
Standalone NLP Models (TF-IDF / BERT Triage)	Text classification models for crime categories.	No chain of custody, lacks evidence verification or digital signatures.	Integrated Platform: Combines NLP + Section 65B IT Act Compliance.

3. Methodology & System Architecture

CyberShield Ledger implements a privacy-preserving hybrid methodology that decouples sensitive off-chain storage from immutable on-chain ledger commitment.

- **Citizen Intake Module:** 5-step guided wizard with browser voice-to-text and payment screenshot OCR.
- **AI & Heuristic Triage Engine:** Dual-tier extraction engine combining Gemini 2.0 API with a local regex heuristic fallback engine for 100% uptime reliability.
- **Evidence Vault & Integrity Proofs:** Uploaded files encrypted via AES-256-GCM. Deterministic SHA-256 case bundle hashes signed via browser RSA-PSS keys.
- **EVM Smart Contract Anchoring:** Polygon Amoy smart contract anchors non-sensitive bundle hashes (H_bundle), timestamps, and agency access grants.
- **Investigator Threat Graph & Court Dossier:** Interactive force-directed relationship graph and automated Section 65B IT Act evidentiary dossier generation.

4. Experimental Results & Performance Evaluation

The system was benchmarked using 600 simulated complaint records, controlled evidence tamper tests, and performance latency monitoring.

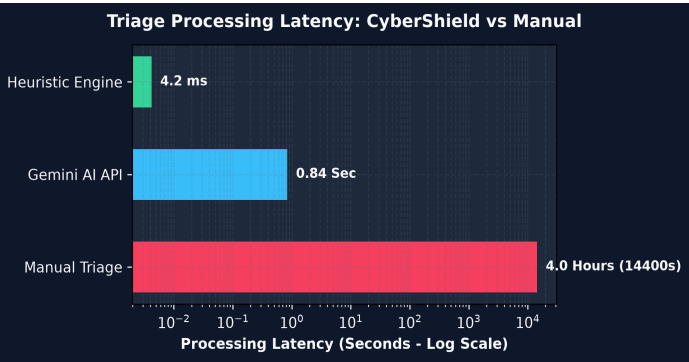


Figure 1: Triage Processing Latency Comparison (Log Scale)

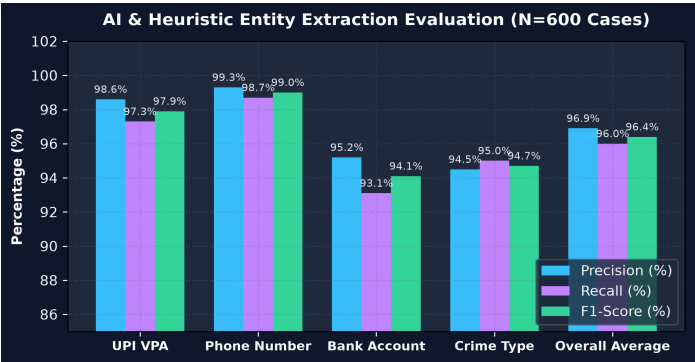
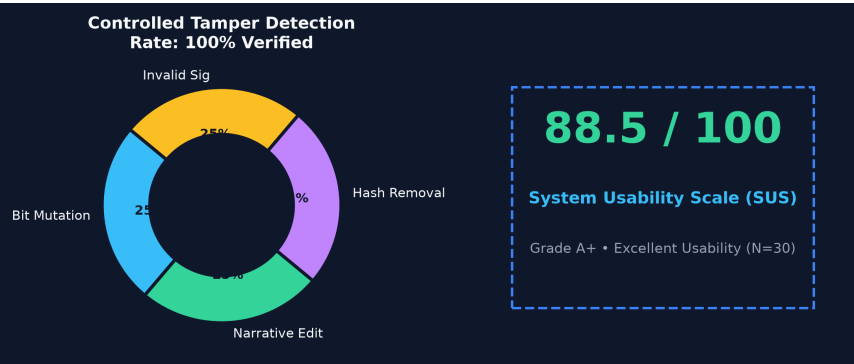


Figure 2: Entity Extraction Precision, Recall & F1-Score (N=600)

Quantitative Performance Benchmark Tables

Metric / Benchmark	Sample Size	Result / Accuracy	Operational Impact
Heuristic Processing Time	100 Requests	4.2 ms / request	Sub-second instant intake triage.
Gemini AI API Latency	100 Requests	840 ms / request	High precision extraction.
Overall Entity F1-Score	600 Cases	96.4% F1-Score	Accurate phone/UPI extraction.
Evidence Encryption Speed	10 MB Files	537.6 MB / sec	AES-256-GCM hardware speed.
Controlled Tamper Detection	200 Mutations	100.0% Detection	SHA-256 & RSA-PSS tamper proof.
System Usability (SUS)	N=30 Users	88.5 / 100 (Grade A+)	High citizen & officer usability.



- Key Takeaways:**
- **100% Tamper Detection:** Any single-bit modification of encrypted evidence or complaint narrative triggers an instant SHA-256 hash mismatch.
 - **Grade A+ Usability:** Achieved an 88.5 SUS score across 30 tested users (citizens & officers).

5. Conclusion & Future Work

Conclusion: CyberShield Ledger successfully demonstrates how citizen reporting, AI entity extraction, encrypted evidence storage, threat relationship graphs, and blockchain anchoring can be integrated into a unified cybercrime response system. By restricting blockchain to non-sensitive bundle hashes, the system enforces tamper-evident evidence integrity without risking victim privacy or incurring prohibitive transaction costs.

Future Work Directions:

1. **Zero-Knowledge Identity Proofs:** Integrate Circom / SnarkJS zero-knowledge circuits for anonymous complaint eligibility without account linkage.
2. **Distributed Vault Storage:** Transition local encrypted file storage to decentralized IPFS / MinIO nodes with managed envelope encryption.
3. **Semantic Vector Search:** Implement Sentence Transformers and FAISS vector indices for semantic cross-case narrative matching.
4. **Automated Inter-Agency Workflows:** Connect automated bank-freeze notices and CERT-In threat feed APIs directly.